ZCAS UNIVERSITY

BSCSE32 - BACHELOR OF SCIENCE IN SOFTWARE ENGINEERING

CCS3242 SOFTWARE PROJECT WITH AGILE TECHNIQUES

FINAL EXAMINATIONS

MONDAY, 2ND DECEMBER, 2024

TIME: 09:00 – 12:00 HOURS

TIME ALLOWED: THREE (03) HOURS

INSTRUCTIONS TO CANDIDATES:

1. **SECTION A:** This section is compulsory. You must answer **Question 1.**
2. **SECTION B:** Attempt **any three (03)** questions out of the **four (04)** questions provided.
3. Time allowed: **3 Hours plus five minutes to read through the paper.**
4. This examination paper carries a total of **100 marks.**
5. Candidates **MUST NOT** turn this page **until** the Invigilator tells them to do so.

SECTION A – COMPULSORY QUESTIONS

INSTRUCTIONS: Section A **must** be attempted by all candidates. You must answer **Question 1** that follows the case study provided.

CASE STORY: "PUNZILANI UNIVERSITY LIBRARY"

Early this year, the University Resource Manager at ***Punzilani University*** presented her need for a new system to automate the processes and functions at the University Library to the Software Engineer.

"We need a system that allows us to manage our library. It needs to let us add, remove and manipulate books, as well as add, remove and manipulate customer details. It should keep a database of all the books that are available on the shelves and those that are in the storeroom. Patrons should be able to view our catalogue through a webpage and place holds on the books that they wish to reserve." She narrates.

Assume you are the Software Engineer at the above university;

QUESTION 1

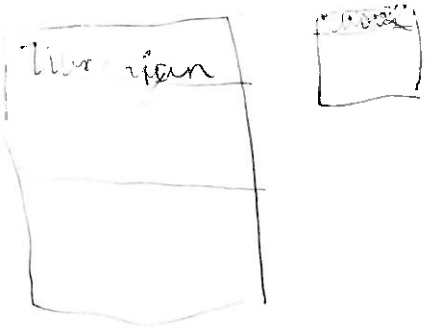
a) Explain the steps involved in Natural Language Analysis (NLA) and how it aids in identifying candidate classes and methods. **(10 marks)**

b) Create a class diagram for the New Library System at ***Punzilani University***. Be certain to indicate identifiers and cardinality constraints. **(20 marks)**

c) Discuss the criteria you should use to decide whether a class can be included in a class diagram or excluded as irrelevant. **(04 marks)**

d) List (02) TWO factors would influence your choice of implementation language for the system? Justify your choice. **(06 marks)**

(Total marks for Section A: 40 marks)



SECTION B

CHOOSE ANY THREE (03) QUESTIONS

QUESTION 2

CASE STORY: ***TechZed Limited***

A medium-sized Zambian company, TechZed Limited, decided to use Scrum for their new mobile application development project. The team is divided into smaller sub-teams, with one Scrum Master overseeing the project. Every two weeks, they hold Sprint Planning meetings, daily stand-ups, and Sprint Reviews. However, the Product Owner is struggling to prioritize the product features effectively, which is causing delays.

a) Describe the main differences between Kanban and Scrum. **(04 marks)**

b) Explain the concept of a Sprint, the Sprint Backlog and Product Backlog in Agile.

(06 marks)

c) Discuss the role of the Product Owner and of the Scrum Master in an Agile project.

(04 marks)

d) Describe static and dynamic models in software design and explain their purpose.

(06 marks)

(Total: 20 marks)

QUESTION 3

CASE STORY: ***AutoDrive Advanced Driver Assistance Systems (ADAS)***

An automotive company, AutoDrive, has decided to implement Agile to improve the development of its Advanced Driver Assistance Systems (ADAS). The project involves a cross-functional team working on features such as automated parking and collision avoidance. The product requirements are subject to change based on safety regulations and market demands, and the team needs to react quickly to these changes.

a) Compare and contrast Agile and Waterfall methodologies. **(10 marks)**

b) Explain how Agile can be applied in the development of Advanced Driver Assistance Systems (ADAS). **(05 marks)**

c) Describe how Agile can be applied in the development of an AI-powered healthcare system.

(05 marks)

(Total: 20 Marks)

QUESTION 4

CASE STORY: *GRZ government project*

A large GRZ government project is being developed to track environmental changes using IoT sensors across different regions. The project is complicated by changing governmental policies, weather conditions, and real-time data requirements. The team is considering using Agile to better respond to these changing variables, but there are concerns about how to manage risks and handle unexpected changes in scope.

a) Describe how Agile approaches risk management. **(5 marks)**

b) Explain how the team can identify, assess, and mitigate risks throughout the project lifecycle using Agile principles. **(5 marks)**

c) Describe how Agile promotes continuous improvement. **(5 marks)**

d) Explain the process the team would use to integrate regular retrospectives to address project challenges and improve the development process throughout the IoT sensor project. **(5 marks)**

(Total: 20 marks)

QUESTION 5

a) Describe how the below techniques can be used in an Agile (DSDM) project: Modelling

i) **MoSCoW** **(4 marks)**

ii) **Prioritization** **(4 marks)**

iii) **Facilitated Workshops** **(4 marks)**

b) Describe the following Scaling Frameworks in relation to Scaled Agile methods :

i) **SAFe** **(4 marks)**

ii) **LeSS** **(4 marks)**

(Total: 20 marks)

END OF EXAMINATION PAPER